

TrES-3b

R-2629

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_210603 · created 2026-08-29T00:09:40+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459368.8556 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.193 +/- 0.058
Transit depth (Rp/Rs)^2	3.73 +/- 2.25 [%]
Orbital Inclination (inc)	85.51 +/- 2.92
Airmass coefficient 1 (a1)	1.622 +/- 0.046
Airmass coefficient 2 (a2)	-0.46 +/- 0.55
Scatter in the residuals of the lightcurve fit is	2.8 %
Best Comparison Star	#3 - [307, 285]
Optimal Aperture	2.89
Optimal Annulus	11.87
Transit Duration (day)	0.075 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.193 ± 0.058 vs 0.16309 (deviation 0.52σ)
Duration fit vs published	0.075 d vs 0.0567 d (deviation 1.31σ)
Mid-time O–C	-4.7 min
Depth SNR	3.3
Residual scatter	2.8 %

Light curve

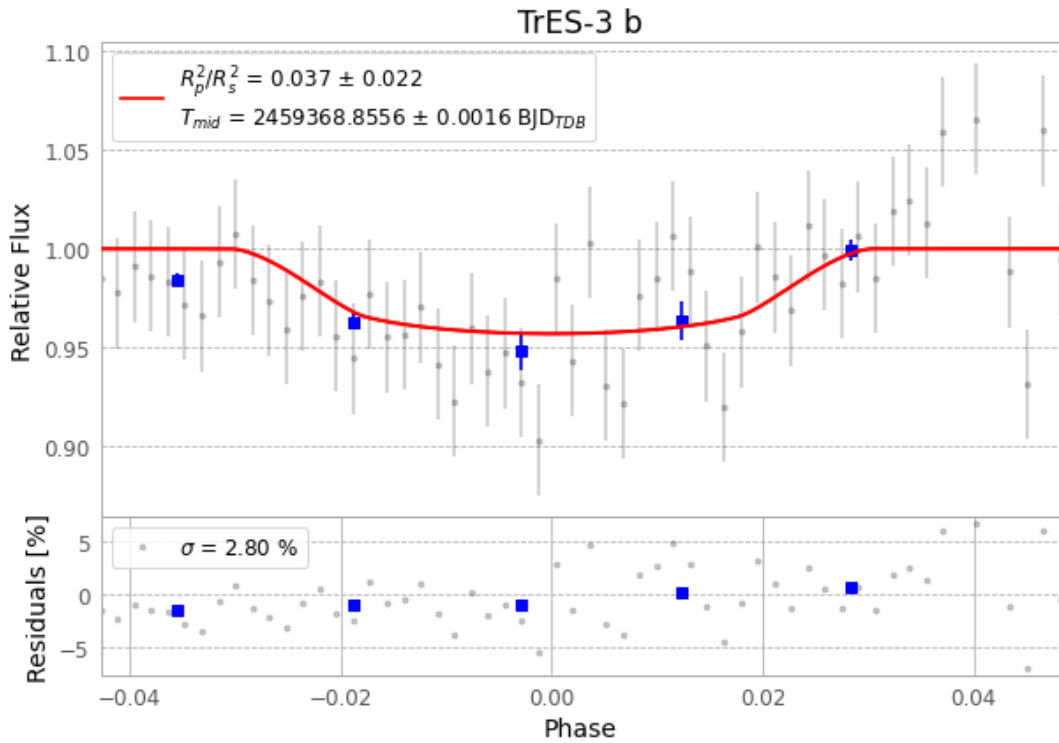


Figure 1 — FinalLightCurve_TrES-3 b_2021-06-03.png (SHA-256 7d76497d951c91a1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [378, 201], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 740612ba4afb9742...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

63 FITS frames (41 MB), TRES-3210603070612.FITS → TRES-3210603101212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-210603.log (0375225b2b5f...), FinalLightCurve_TrES-3 b_2021-06-03.png (7d76497d951c...), FinalLightCurve_TrES-3 b_2021-06-03.csv (8498de08328f...)

Compute resources

Wall time 246.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-29 00:09Z.